ADALINE POWELL

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SUMMARY

Data science and political science researcher who pairs rigorous quantitative methods with social-science research design. Experience spans real-time data engineering (Microsoft Fabric, Azure IoT Hub, Snowflake, KQL/PySpark), predictive modeling and NLP, Power BI/DAX analytics, and multi-agent AI systems built directly on the Anthropic SDK. Currently completing an MS in Data Science and Applied AI at UW-Milwaukee, building on graduate training in political science research methodology.

EDUCATION

Master of Science, Data Science and Applied AI — *Expected Dec. 2026*
University of Wisconsin–Milwaukee

Master of Arts, Political Science — *2025*
University of Wisconsin–Milwaukee

Bachelor of Arts, Political Science (Pre-Law) — *2023*
University of Wisconsin–Milwaukee

Bachelor of Arts, Philosophy — *2023*
University of Wisconsin–Milwaukee

WORK EXPERIENCE

Data Science Intern, Advanced Analytics — *June – Aug. 2026*
Delta Defense, LLC · West Bend, WI · Event Benchmarking & Predictive Pipeline (USCCA)

- Built an attendee-focused event benchmarking application that forecasts attendance and membership conversion for upcoming events, replacing reporting that could only evaluate events after the fact with no modeled expectation to explain under- or over-performance; now used in recurring weekly leadership meetings.
- Modeled registrant-level attendance forecasting, an AE-reported-vs-system-check-in discrepancy model, and a lead-to-membership conversion model in Python (pandas, scikit-learn, statsmodels), with data extraction and transformation in Snowflake SQL and dbt.
- Trained on an 800K+ row dataset assembled from Snowflake event, registrant, AE-report, and membership tables; caught and corrected a data-quality error in an AI-generated query pulling the wrong attendance field, which directly motivated the registrant-level redesign.

Research Analyst — *May – Aug. 2026*
Wisconsin Women's Council · WWC Status of Women Dashboard

- Designed a county-level Power BI dashboard on the status of women in Wisconsin — demographics, earnings, economic security, health, and domestic violence across all 72 counties — creating a single reliable view where the data had been siloed across four government sources with inconsistent structure and undocumented quality issues.
- Architected a unified star schema with a shared conformed geography dimension, built Python/pandas ingestion pipelines, and authored and audited 90+ DAX measures; designed a multi-agent Claude/VS Code validation system with YAML semantic models for ongoing pipeline QA.
- Integrated Census ACS 5-year API data (~75,000 rows), FBI NIBRS domestic violence incident data, and Wisconsin DHS WISH birth records; identified and resolved 20+ data-quality defects (e.g., a silently corrupting Census sentinel value, a 0%-match geographic join), verifying every fix against primary-source data.

PROJECTS

Nonprofit Financial Analysis Agent — *2026*
Argosy Foundation · Python, Anthropic SDK, Excel

- Automated nonprofit financial review with a Python agent that reads IRS Form 990 PDFs — including scanned filings — into structured, auditable data with computed health ratios, replacing manual review of 20–40 dense pages per filing that couldn't scale across organizations and years.
- Developed on the Anthropic SDK (~1,000 lines of Python) with a line-item extraction contract anchoring each of 40+ fields to its exact IRS form line, a ratio engine computing 18 metrics against sector benchmarks, and an idempotent openpyxl Excel pipeline with automatic change-detection.
- Processed 63 filings across 16 organizations spanning tax years 2018–2025; cross-checked against an independent dataset, traced a systematic error to a 2018 accounting-standard change affecting 21 of 63 filings, and reprocessed the full dataset.

Discovery World Hackathon — *Jan. 2026*
Microsoft Fabric, Azure IoT Hub, KQL, PySpark

- Engineered a real-time streaming pipeline and interactive exhibit dashboard for industrial conveyor-belt systems that previously had no live visibility or fault forecasting, and no accessible way to make that data engaging for a public, kid-facing audience.
- Served as primary data engineer, designing the ELT flow on Microsoft Fabric (Eventhouse, Eventstreams) with Azure IoT Hub, KQL, and PySpark across a Bronze/Silver/Gold medallion architecture, plus predictive maintenance models to detect anomalies and forecast equipment faults.

- Processed high-velocity streaming IoT sensor telemetry from live manufacturing equipment for real-time analytics.

UWM Track & Field Analytics Platform *Spring 2026*

Python, Flask, SQLite, scikit-learn, XGBoost, Plotly

- Vibecoded a full-stack, role-based analytics platform centralizing athlete competition, testing, and practice data and predicting season PRs — replacing three disconnected sources (TFRRS, coach spreadsheets, paper logs) that made connecting training indicators to outcomes impossible.
- Implemented the platform in Flask/SQLite with Flask-Login role guards, a requests/BeautifulSoup scraper automating TFRRS collection, per-event Random Forest and XGBoost models (scikit-learn, joblib), and interactive Plotly dashboards.
- Unified TFRRS competition history for ~154 UWM athletes with six seasons of multi-format Excel fall-testing data and coach-provided practice logs into a six-table relational schema.

TECHNICAL SKILLS

Data Engineering & Platforms: ELT Pipelines, Streaming Data, Real-Time Analytics, SQL, Snowflake, dbt, Databricks, Microsoft Fabric (Eventhouse, Eventstreams), Azure IoT Hub, KQL, PySpark, ArcGIS

Data Science & Modeling: Machine Learning, NLP & Topic Modeling, Regression (Logistic, Negative Binomial), Predictive Modeling, Feature Engineering, Cross-Validation, Random Forests, XGBoost, Neural Networks

BI & Analytics: Power BI (DAX), Sigma, Streamlit, Data Visualization

Agentic AI Development: Anthropic SDK/API, Multi-Agent System Design, Prompt Engineering, Claude (VS Code / Claude Code)

Programming & Tools: Python (pandas, scikit-learn, Flask, BeautifulSoup), R, SQL, SPSS, Qualtrics

Certificates: Quantitative Social Data Analysis

EXTRACURRICULARS

Northwestern Mutual Data Science Institute

Women's Track and Field